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RE: Digital Lending Portfolio — Risk Segmentation, Early Warning Model, and Deployment Recommendation
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1. Executive Summary

This memo presents findings from a risk segmentation and early-warning model built on a 35,000-loan synthetic digital lending portfolio, constructed to reflect the risk dynamics of an emerging-market unsecured/BNPL/SME lending book. The objective was to answer five questions: which segments carry materially different risk, how acquisition channels affect unit economics, which products balance growth and risk, how pricing should be tailored by segment, and what the CRO should monitor going forward.

Three findings drive this memo's recommendation:

- A four-segment risk structure exists in the portfolio, with default rates ranging from 2.2% (Prime Stable) to 14.9% (Overleveraged) - a gap wide enough to justify differentiated pricing and underwriting treatment.
- An XGBoost/CatBoost early-warning model achieves a 0.71 out-of-sample ROC-AUC - close to this portfolio's estimated theoretical ceiling of 0.72, meaning the model captures nearly all extractable signal given inherent repayment randomness. We do not claim higher accuracy than the data supports.
- Critically, simulating the model as an automatic approve/decline gate destroys portfolio value at every threshold tested (estimated ₹38.6M net value destruction on the test portfolio) because it forgoes interest on the large majority of flagged loans that are, in fact, good. Routing flagged loans to manual underwriter review instead reverses this to an estimated ₹50.1M net benefit.

Recommendation: deploy the model as a manual-review trigger, not an auto-decline gate, with segment-differentiated pricing for the two highest-risk segments.

2. Risk Segmentation Findings

Customers were segmented using K-Means clustering on five behavioural signals: cash flow consistency, balance volatility, credit quality indicator, spending shock incidence, and partial payment ratio. The number of clusters (k=4) was selected using silhouette score analysis rather than assumed in advance k=4 produced the highest silhouette score (0.407) among k=2 through k=6 tested.

Segment	Size	Avg Credit Score	Avg Balance Volatility	Default Rate
Prime Stable	16,497 (47.1%)	706	0.254	2.2%
Emerging Affluent	4,041 (11.5%)	691	0.273	5.5%
Volatile Spenders	12,047 (34.4%)	617	0.430	5.6%
Overleveraged	2,415 (6.9%)	591	0.451	14.9%

Source: K-Means clustering (k=4, silhouette-validated), n=35,000. Default rates are empirical, not model-predicted.

The Overleveraged segment (6.9% of the book) carries nearly 7x the default rate of Prime Stable, despite being a minority of volume - this segment is the highest-priority target for both pricing correction and tightened underwriting.

3. Early Warning Model & Calibration

An XGBoost and CatBoost classifier were benchmarked head-to-head. CatBoost was selected as the champion model (0.706 vs. 0.678 out-of-sample ROC-AUC). Both scores sit close to the dataset's estimated noise ceiling of 0.716 - derived from scoring the underlying latent risk signal directly against outcomes - the model is extracting close to the maximum available signal rather than underperforming.

A calibration check was run before trusting the model's probability outputs for any pricing decision. The initial model showed a substantial calibration gap (0.345 average absolute deviation between predicted and actual default rates) - a direct consequence of the class-imbalance correction (scale_pos_weight) used during training, which improves ranking but distorts probability magnitude. This was corrected using isotonic regression fit on a held-out calibration split, reducing the average gap to 0.006.

This step matters operationally: any segment-level pricing or capital allocation decision built on the model's raw, uncalibrated probabilities would have systematically overstated risk by 30+ percentage points in some bands. We flag this because it is a common and easily-missed failure mode in imbalanced credit models, and recommend any future model refresh include a calibration check as a standard release gate.

SHAP analysis confirms the model is learning the correct risk structure: partial payment ratio is the single largest driver of predicted risk (mean |SHAP| = 0.359, strongly positive direction), followed by credit quality indicator and monthly income - consistent with the underlying economic logic of the portfolio.

4. Acquisition Channel & Product Economics

Brief Q2 asks how acquisition channels affect portfolio quality and unit economics. The table below ranks all four channels by average value proxy per customer (interest income minus expected loss minus acquisition cost):

Channel	Default Rate	Avg CAC	Avg Turnaround	Avg Value/Customer
Organic Search	4.0%	₹50	12 hrs	₹19,163
Bank Referral	3.0%	₹300	48 hrs	₹18,818
Performance Ads	5.0%	₹1,200	4 hrs	₹15,906
Aggregator Platform	6.0%	₹850	2 hrs	₹15,062

Source: Cell 5 channel economics analysis, n=35,000.

The pattern is consistent with adverse selection: channels with the fastest, lowest-friction approval (Aggregator Platform, Performance Ads) show both higher default rates and lower

value-per-customer than slower, more deliberate channels (Bank Referral, Organic Search) - despite Aggregator Platform driving the largest origination volume (12,263 of 35,000 loans, 35% of the book). Speed-to-approval is a growth lever, but this data suggests it is currently being purchased at a real risk cost on this channel specifically.

Brief Q3 asks which products and ticket sizes balance growth against risk. Segmenting by product type and ticket-size tercile shows High -Ticket Unsecured Personal Loans and High-Ticket SME Working Capital loans deliver the highest average value proxy ($\approx ₹36,700$ - $36,800$ per loan) - substantially higher than BNPL / Credit Line products ($₹1,138$ - $2,654$), which carry the smallest ticket sizes by design and offer correspondingly thin unit economics per loan, though they may still serve a strategic role in customer acquisition and upsell pathways not captured in this single-loan view.

5. Leadership Monitoring: Recommended KPIs

Brief Q5 asks what performance metrics senior leadership should monitor to proactively manage risk and growth. Based on the findings above, we recommend the following as standing dashboard metrics, reviewed monthly:

- Segment mix shift - % of new originations falling into the Overleveraged and Volatile Spenders segments. A rising share signals underwriting or acquisition drift toward higher-risk borrowers before default rates themselves move.
- Model calibration drift - re-run the calibration check (Section 3) on a rolling quarterly basis. A widening gap between predicted and actual default rates is an early signal that the model needs retraining, independent of overall AUC.
- Channel-level value proxy trend - track average value-per-customer by acquisition channel monthly, not just volume or CAC in isolation. A channel can grow volume while quietly destroying unit economics, as seen with Aggregator Platform in this analysis.
- Manual review queue conversion rate - % of flagged loans the review team ultimately approves vs. declines. This validates (or invalidates) the 70%/85% planning assumptions behind the ₹50.1M projected benefit, and should be the first metric checked after any pilot.
- Pricing gap by segment - required yield minus current average rate, by segment (Section 2 methodology). This converts segmentation into an actionable, monitorable pricing signal rather than a one-time analysis.

6. Deployment Recommendation & Projected Impact

The brief's evaluation criteria require heavily penalizing missed defaults. We optimized the decision threshold using an F2 score (recall weighted 2x over precision) rather than a balanced F1 score, reflecting this priority. At the resulting threshold, the model flags loans for review at a rate that catches a meaningfully higher share of true defaults than a naive cutoff, at the cost of a higher false-positive (over-flagging) rate.

We then simulated two deployment models on the test portfolio (7,000 loans):

Deployment Model	Mechanism	Net Portfolio Impact
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Auto-Decline	Flagged loans automatically rejected	- ₹38.6M (value destroyed)
Manual Review (Recommended)	Flagged loans routed to underwriter; good loans re-approved, bad loans declined	+₹50.1M (net benefit)

Manual review economics assume: ₹150/loan review cost, 70% reviewer catch-rate on truly bad flagged loans, 85% correct re-approval rate on truly good flagged loans. These rates are illustrative planning assumptions, not measured data, and should be validated with a small pilot cohort before full rollout.

The reason auto-decline fails is structural, not a tuning error: at any threshold sensitive enough to catch a useful share of defaults, the large majority of flagged loans are still good loans (in our test portfolio, 1,875 of 2,054 flagged loans were non-defaulting). Auto-declining all of them forfeits far more interest income than the avoided losses justify. Manual review recovers most of that forfeited income while still removing genuine bad loans before disbursement.

7. Recommended Actions

- Deploy the calibrated CatBoost model as a manual-review trigger at the F2 - optimal threshold - not as an auto-decline gate.
- Apply segment-differentiated pricing: Overleveraged and Volatile Spenders segments should be repriced toward their risk-adjusted required yield; Prime Stable and Emerging Affluent segments are currently priced above required yield and represent volume-growth opportunity.
- Re-route Aggregator Platform originations through additional verification steps - this channel shows the lowest value-per-customer of the four acquisition channels despite high volume, consistent with adverse selection from fast, low-friction approval.
- Re-run the calibration check on any model refresh - this is now a known failure mode for this model family and should be a standing release gate, not a one-time fix.
- Pilot the manual - review assumptions (review cost, catch rate, approval rate) with a small live cohort before committing to portfolio-wide rollout, since these are currently planning estimates.

8. Limitations & Caveats

- This analysis uses a synthetic dataset constructed to reflect realistic lending dynamics per the project's data design guidelines; all figures are illustrative of methodology, not live portfolio figures, and require revalidation on real origination data before any production decision.
- The 0.71 AUC ceiling reflects irreducible randomness deliberately built into the simulated repayment behaviour, mirroring real-world unpredictability; this should not be read as a deficiency in modelling approach.
- Manual review catch/approval rates are assumptions, not measured outcomes, and are the single most important input to validate before relying on the ₹50.1M projected benefit.